

Automatic Image Analysis Script

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1 Overview

This script collects notes for the Automatic Image Analysis material. It is intended as a working LaTeX document that can be expanded with lecture notes, exercise solutions, definitions, and project observations.

2 Core Topics

2.1 Image Formation and Representation

Add notes on image acquisition, sampling, color spaces, filtering, and basic image transformations.

2.2 Learning-Based Methods

Add notes on neural networks, model training, evaluation, and uncertainty.

2.3 Visual Object Models

Add notes on feature representations, object models, detection, and recognition.

2.4 Probabilistic Estimation

Add notes on Bayesian reasoning, probabilistic models, and parameter estimation.

3 Exercises

Use this section for worked exercise notes and derivations.